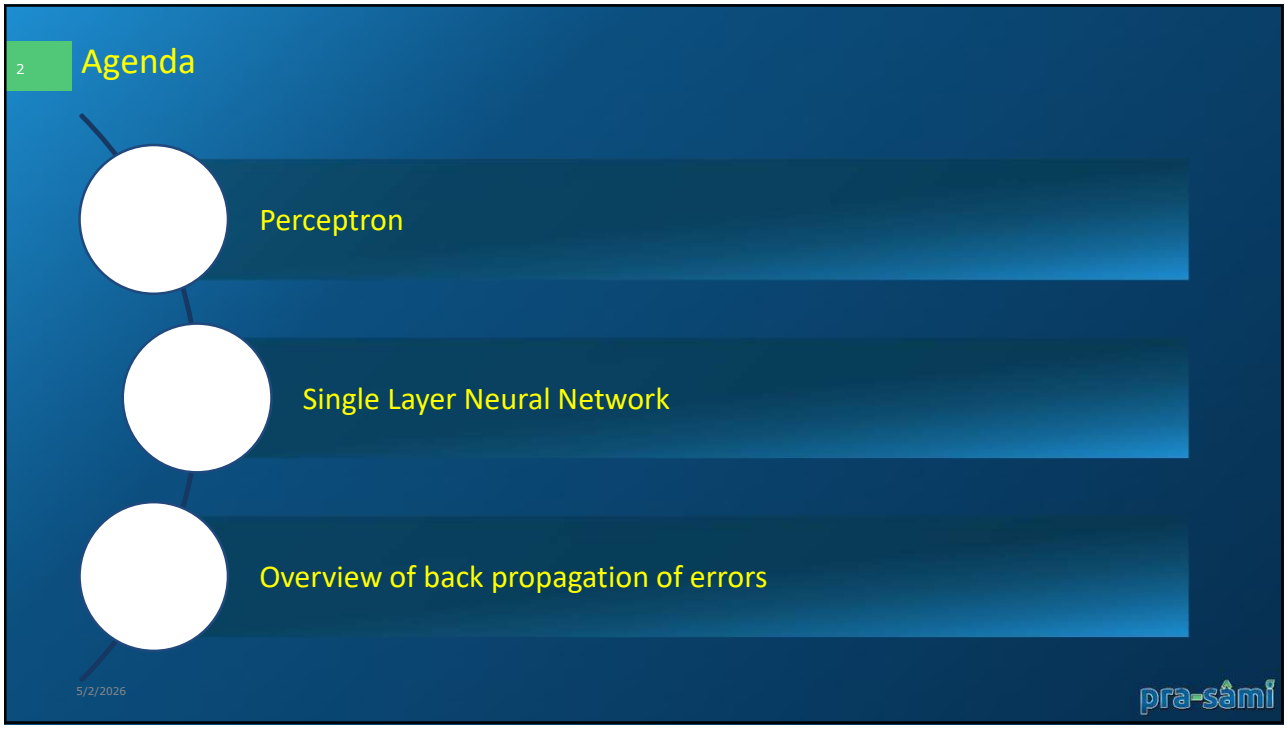
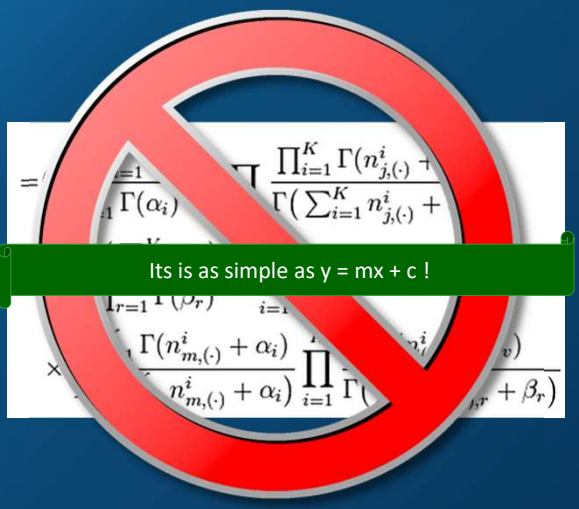

Perceptron, ANN, Back-Prop
Deep Neural Networks
Session 02
Pramod Sharma
pramod.sharma@prasami.com


2
Agenda
Perceptron
Single Layer Neural Network
Overview of back propagation of errors
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3 Solution to Equation of Perceptron



Its is as simple as y = mx + c !



Frank Rosenblatt

5/2/2026

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4 From MP Neuron → Perceptron

McCulloch–Pitts Neuron	Perceptron
❑ Binary inputs: $\{0, 1\}$	❑ Inputs are numeric (not just binary)
❑ No learning (fixed behavior)	❑ Each input has a learnable weight
❑ Threshold-based decision	❑ Computes weighted sum + bias ❖ $z = w^T x + b$
❑ Inhibitory inputs have veto power	❑ Learns from data (training algorithm)

Key Shift

- ❑ From fixed rules → learnable model
- ❑ From binary logic → weighted computation
- ❑ From threshold only → bias + weights

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5

To Play or not to Play…

In millimeter | In °C | completed? – 0 : No; 1: Yes | How many | Available? | Rupees/hour

id	Rains	Temp	Homework	Team Members	Equipment	Ground	Played
1	0	38	1	15	0	600	1
2	0	25	1	15	1	800	1
3	0	26	1	15	1	1000	1
4	5	27	1	10	1	600	0
5	20	23	0	8	1	1800	0
6	30	22	0	6	0	600	0

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6

Weights and Scaling

- Different features do not contribute equally
- Assign weights to capture feature importance
- Values of each features are in different order of magnitude
  - Summation is not going to work, Large-scale features can dominate the model
  - Scale the features between 0 and 1

id	Rains	Temp	Homework	Team Members	Equipment	Ground	Played
1	0	38	1	15	0	600	1
2	0	25	1	15	1	800	1
3	0	26	1	15	1	1000	1
4	5	27	1	10	1	600	0
5	20	23	0	8	1	1800	0
6	30	22	0	6	0	600	0

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Direction of Impact

- Features can influence output differently
- Positive impact → higher value increases output
- Negative impact → higher value decreases output
- Examples:
  - Team members → positive contribution
  - Ground cost → negative contribution

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Perceptron

- In MP Neuron Model,
  - All inputs had same weights
  - Threshold 'w_0' could take limited values
  - Every feature needed to be [0,1]
- Perceptron model introduced different weights to different inputs features and bias
- Real values are also accepted
  - Temperatures are in tens and ground rent is in hundreds.
  - Min – Max – Scaler to compensate for huge difference is values
- Threshold 'w_0' can take any value
- Outputs are still [0, 1]

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Perceptron

- Loss Function:
  - A correction is applied on the outputs
  - To adjust values of 'w_i' to reach right results
  - It would also give us indications of what weights to be fixed to arrive at the solution

- Activation function g(x) is applied as follows:
  - If $\sum x_i . w_i \geq w_0$ => $\hat{y} = 1$
  - If $\sum x_i . w_i < w_0$ => $\hat{y} = 0$

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Perceptron – Data Preprocessing

- Scaled Data (all columns to be between 0 and 1)

id	Rains	Temp	Homework	Team Members	Equipment	Ground	Played
1	0.00	0.00	1.00	1.00	0.00	1.00	1
2	0.00	0.81	1.00	1.00	1.00	0.83	1
3	0.00	0.75	1.00	1.00	1.00	0.67	1
4	-0.17	0.69	1.00	0.44	1.00	1.00	0
5	-0.67	0.94	0.00	0.22	1.00	0.00	0
6	-1.00	1.00	0.00	0.00	0.00	1.00	0

- What about reverse correlation

- Two option to address reverse correlation
  - Take negative of values
  - Use negative weight

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Perceptron – Data Preprocessing

- Consider "Ground" and "Team Members" as features
- Associate weights

id	Rains	Temp	Homework	Team Members	Equipment	Ground	Played
1	0	38	1	15	0	600	1
2	0	25	1	15	1	800	1
3	0	26	1	15	1	1000	1
4	5	27	1	10	1	600	0
5	20	23	0	8	1	1800	0
6	30	22	0	6	0	600	0

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Perceptron – Weights

- Weights determine the influence of each feature on the output
  - Higher weight → stronger impact on prediction
- Initialization of Weights
  - Initially, weights are assigned small random values

id	Threshold	Team Members		Ground		Calculations	Likely	Played	Loss
	w0	x1	w1	x2	w2	w0+x1*w1+x2*w2	(y_hat)	(y)	(y-y_hat)^2
1	-1.00	1.00	1.10	1.00	1.00	1.10	1	1	0
2	-1.00	1.00	1.10	0.83	1.00	0.93	1	1	0
3	-1.00	1.00	1.10	0.67	1.00	0.77	1	1	0
4	-1.00	0.44	1.10	1.00	1.00	0.49	1	0	1
5	-1.00	0.22	1.10	0.00	1.00	-0.76	0	0	0
6	-1.00	0.00	1.10	1.00	1.00	0.00	1	0	1

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Perceptron – Weights and Loss

- Our best solution would be where ground truth and predicted values are same
- Loss is some function of ground truth and predicted values
- And we want it to be cumulative, Square of difference looks promising
  - $\ell(\hat{y}, y) = (y - \hat{y})^2$
  - Our overall loss was 2.
- By adjusting weights (w_1, w_2) and threshold (w_0) we can bring the loss to minimum (zero in this case)

id	Threshold	Team Members		Ground		Calculations	Likely	Played	Loss
	w0	x1	w1	x2	w2	w0+x1*w1+x2*w2	(y_hat)	(y)	(y-y_hat)^2
1	-2.00	1.00	1.10	1.00	1.40	0.50	1	1	0
2	-2.00	1.00	1.10	0.83	1.40	0.27	1	1	0
3	-2.00	1.00	1.10	0.67	1.40	0.03	1	1	0
4	-2.00	0.44	1.10	1.00	1.40	-0.11	0	0	0
5	-2.00	0.22	1.10	0.00	1.40	-1.76	0	0	0
6	-2.00	0.00	1.10	1.00	1.40	-0.60	0	0	0

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Perceptron – Weights and Los

- Our best solution would be w… …ou… …h and p… …re same
- Loss is some function of gro… …dicted
- And we want it to be cumulat… …ifferen… …pro…
  - $\ell(\hat{y}, y) = (y - \hat{y})^2$
  - Our overall loss was 2.
- By adjusting weights (w_1, … …$_0$) v… …to minimum (zero in this case)



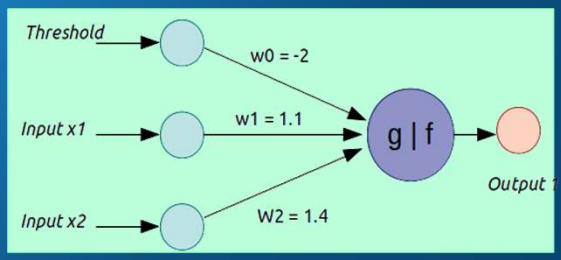
id	Threshold			round		ations		Played	Loss
	w0	x1	1		w2	w1+x	t)	(y)	(y-y_hat)^2
1	-2.00	1.00	1.10	0	1.40		1	1	0
2	-2.00	1.00	1.1	83	1.40		1	1	0
3	-2.00	1.00	1.1	.67	1.40		1	1	0
4	-2.00	0.44	1.1	1.00	1.40		0	0	0
5	-2.00	0.22	1.10	0.00	1.40		0	0	0
6	-2.00	0.00						0	0

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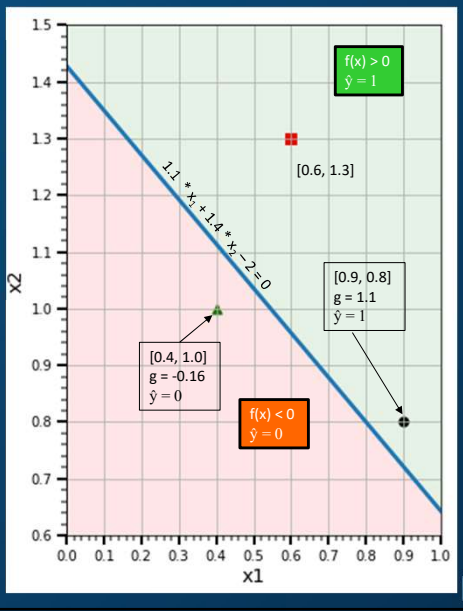
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Perceptron



- We can represent : g = $w_0 + x_1 * w_1 + x_2 * w_2$
  - As g = $[x_1, x_2].\begin{bmatrix} w_1 \\ w_2 \end{bmatrix} + w_0$
- Given: W = $\begin{bmatrix} 1.1 \\ 1.4 \end{bmatrix}$ and w_0 = -2
  - g = $[x_1, x_2].\begin{bmatrix} 1.1 \\ 1.4 \end{bmatrix}$ - 2
  - g = 1.1 * x_1, +1.4 * x_2 - 2



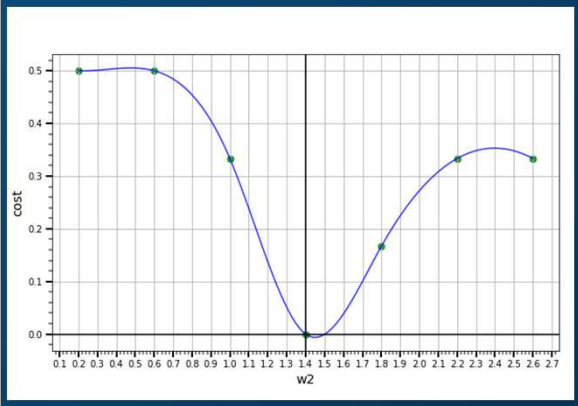
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Perceptron – Adjusting Weights (Optimization Intuition)

- Goal: find w_0, w_1, w_2 that minimize loss.
  - This corresponds to the lowest point on the loss curve
- For illustration:
  - Fix w_0= -2, w_1 = 1.1
  - Vary w_2 0 to 3 to observe effect on loss



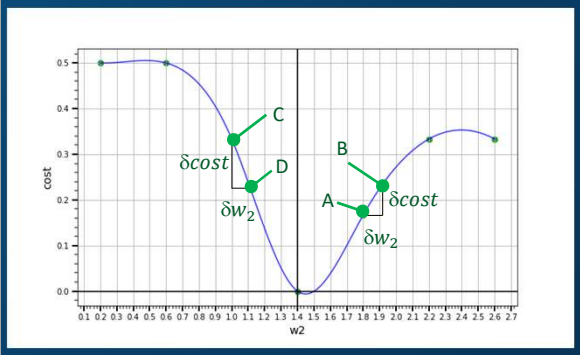
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Perceptron – Understanding the Slope

- The slope (gradient) tells how loss changes with w_2
- Cases:
  - Positive slope
    - Loss increases as w_2 increases → Decrease w_2
  - Negative slope
    - Loss decreases as w_2 increases → Increase w_2
  - Zero slope
    - Minimum point reached



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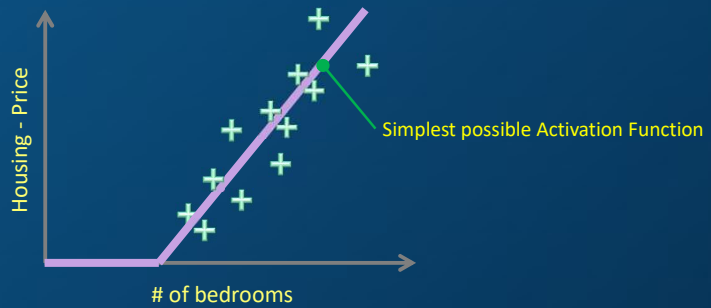
Perceptron – Activation Function

- So we based our entire calculations on:

$$z = w_0 + x_1 * w_1 + x_2 * w_2$$



But that's an equation of straight line!
What happened to all those 'inhibitory' features?

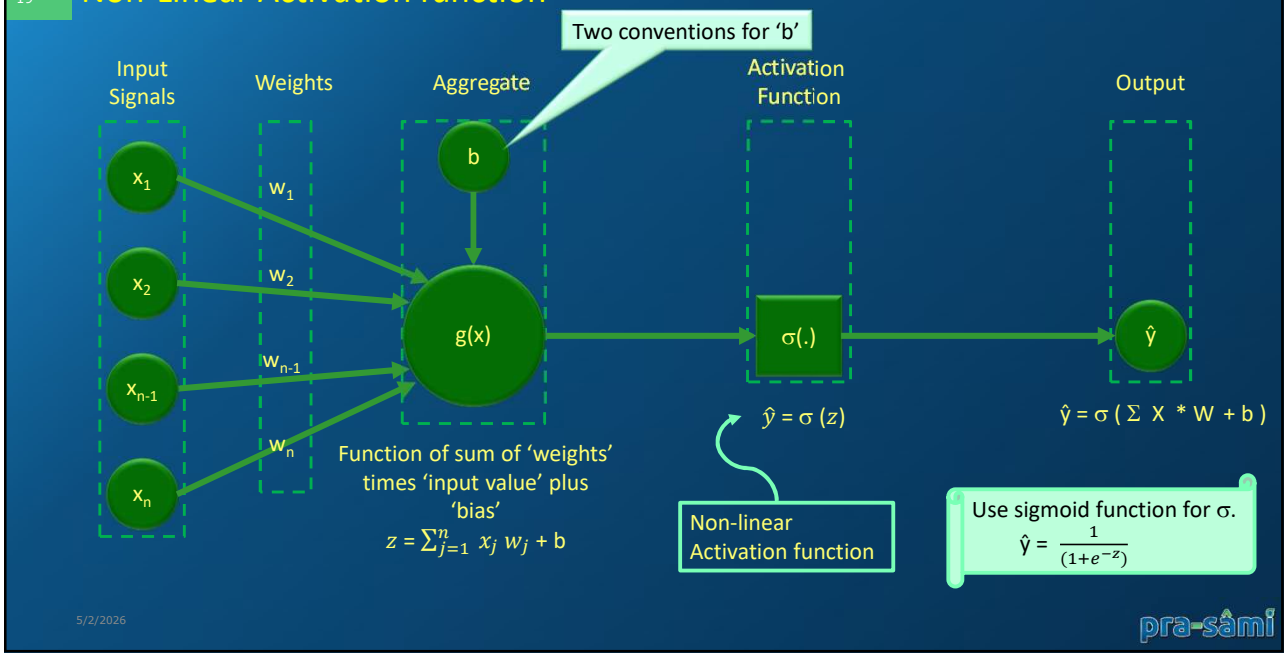


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Non-Linear Activation function

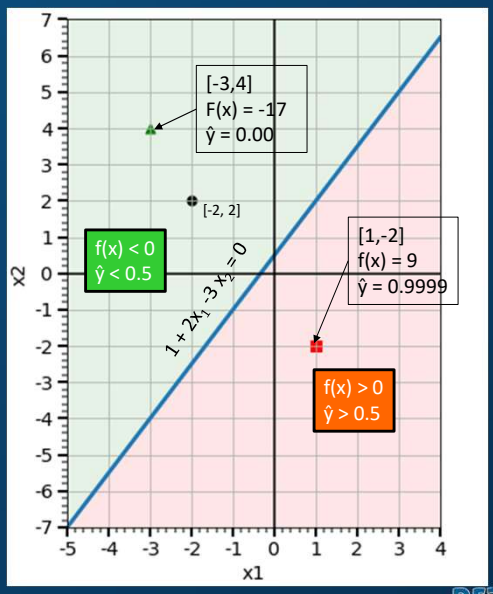


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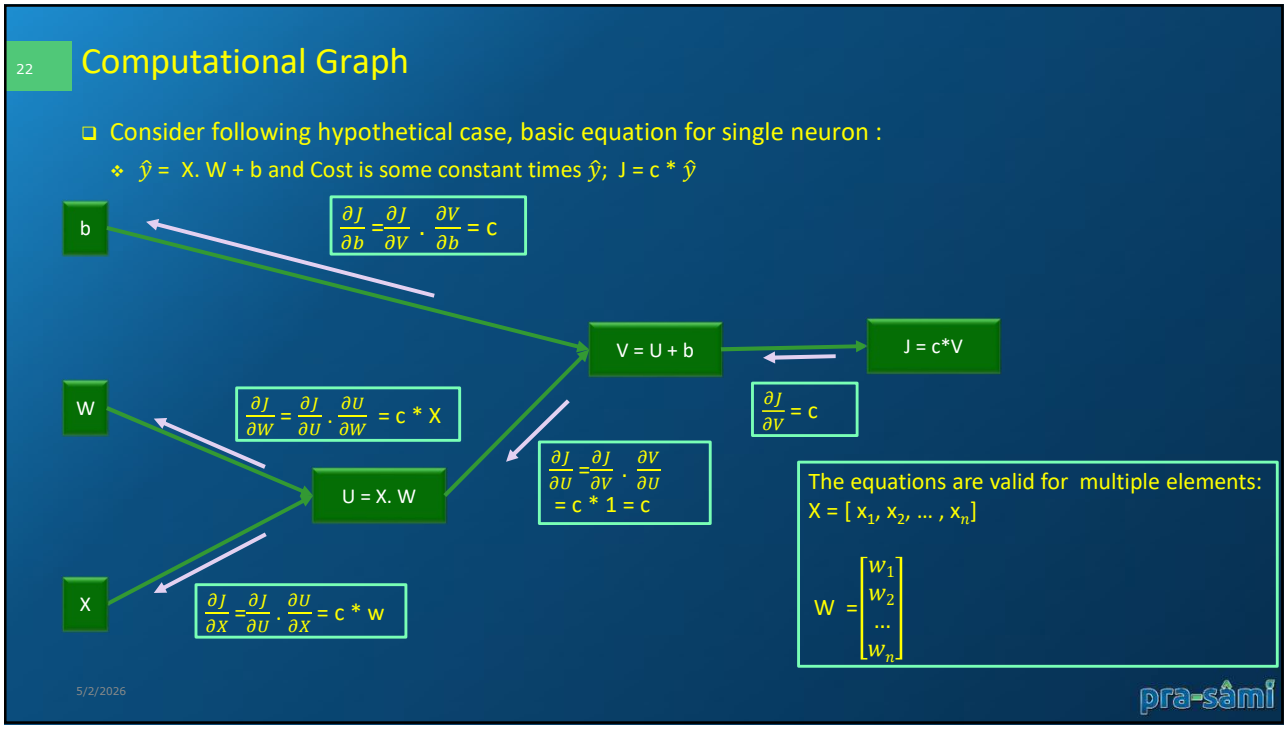
Perceptron with Non-Linear Activation Function

- $\hat{y} = \sigma\ (1 + 2 * x_1 - 3 * x_2)$
- For X = [-3, 4]
  - $\hat{y} = \sigma\ (1 + 2 * (-3) - 3 * 4)$
  - $\hat{y} = \sigma\ (1 - 6 - 12)$
  - $\hat{y} = \sigma\ (-17)$
  - $\hat{y} = 0.0$
- Similarly, for X = [1, -2]
  - $\hat{y} = \sigma\ (1 + 2 * 1 - 3 * (-2))$
  - $\hat{y} = \sigma\ (1 + 2 - 6)$
  - $\hat{y} = \sigma\ (9)$
  - $\hat{y} = 1.0$



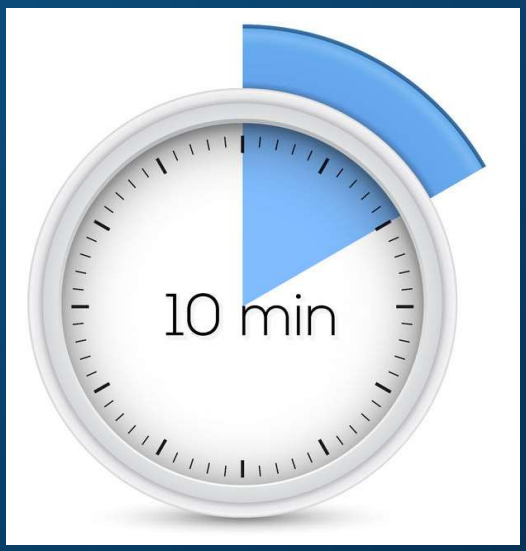
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Perce ith non-linear activation function
ŷ =
For
ŷ =
Are we there yet!
Lets learn some math too!!
Yeehaw!!!
f(x) > 0
ŷ > 0.5
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22
Computational Graph
Consider following hypothetical case, basic equation for single neuron :
$\hat{y}$ = X. W + b and Cost is some constant times $\hat{y}$; J = c * $\hat{y}$
b
$\frac{\partial J}{\partial b} = \frac{\partial J}{\partial V} \cdot \frac{\partial V}{\partial b} = c$
V = U + b
J = c*V
$\frac{\partial J}{\partial V} = c$
W
$\frac{\partial J}{\partial W} = \frac{\partial J}{\partial U} \cdot \frac{\partial U}{\partial W} = c * X$
U = X. W
$\frac{\partial J}{\partial U} = \frac{\partial J}{\partial V} \cdot \frac{\partial V}{\partial U} = c * 1 = c$
X
$\frac{\partial J}{\partial X} = \frac{\partial J}{\partial U} \cdot \frac{\partial U}{\partial X} = c * w$
The equations are valid for multiple elements:
X = [x_1, x_2, … , x_n]
W = $\begin{bmatrix} w_1 \\ w_2 \\ \dots \\ w_n \end{bmatrix}$
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23 Exercise 2 : Computational Graph

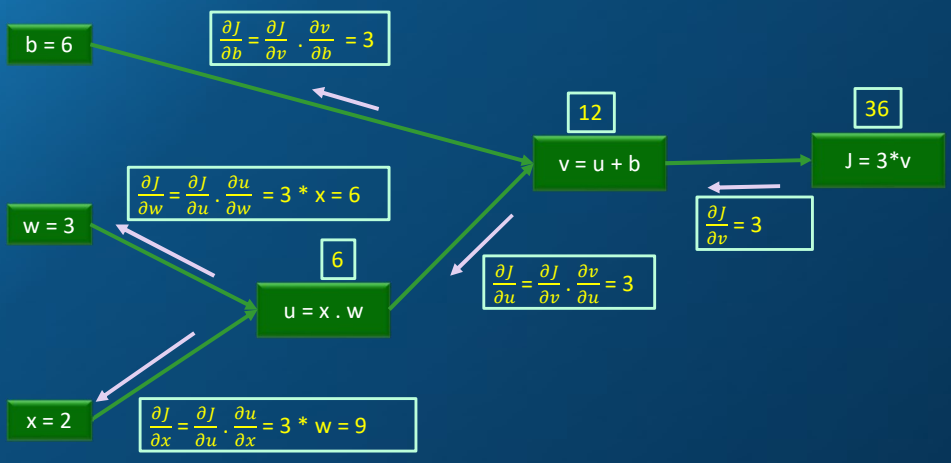
- Given a Cost Function J
  - J (w, x, b) = 3 *(b + x * w)
- Calculate $\frac{\partial J}{\partial W}$, $\frac{\partial J}{\partial x}$ and $\frac{\partial J}{\partial b}$
- Calculate slope at point :
  - b = 6
  - w = 3
  - x = 2



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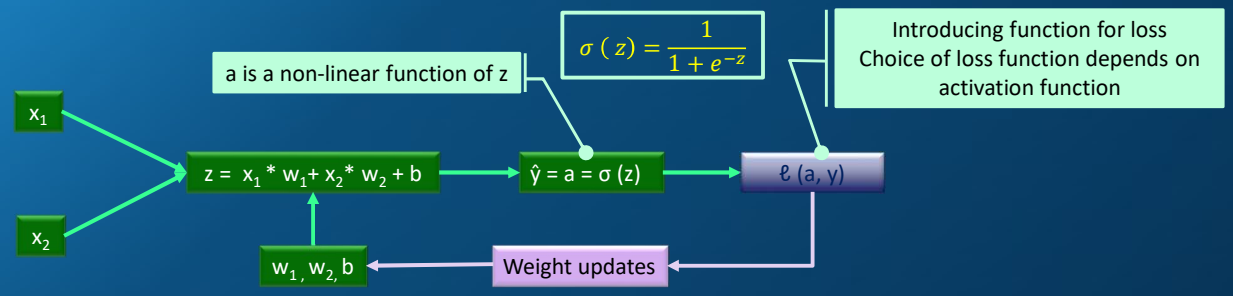
24 Exercise - Solution

- Given a Cost Function J (w, x, b) = 3 *(b + w*x)



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25 Single Path – Maximum Likelihood Estimation (MLE)



Whether for classification or regression, MLE provides a unified principle:

"Optimize model parameters to make the observed data most probable."

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26 Single Path – Maximum Likelihood Estimation (MLE)

- Maximum Likelihood Estimation (MLE) is a statistical technique used to find the best values of model parameters so that the model explains the observed data as well as possible.
- Core Idea:
  - MLE chooses parameters that make the observed (training) data most probable under the model.
- Connection to Neural Networks
  - While training a neural network:
    - We adjust the weights and biases
    - So that the model assigns high probability to correct outputs
    - And low probability to incorrect outputs
- This process is essentially Maximum Likelihood Estimation

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MLE - Classification Tasks

Train the model so that correct classes get the highest probability

- In classification problems, the model outputs a probability distribution over all possible classes for a given input.
- Maximum Likelihood Estimation (MLE) aims to:
  - Find the optimal weights
  - Such that the model assigns the highest probability to the correct (true) class labels

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MLE - Classification Tasks

How is this achieved?

- Instead of directly maximizing likelihood, we:
  - Minimize Cross-Entropy Loss
- Cross-entropy measures the difference between:
  - Predicted probabilities (from the model), and
  - Actual distribution (ground truth labels)
- Lower cross-entropy = better predictions

Take-away

Maximizing likelihood (MLE) Minimizing cross-entropy loss

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MLE - Regression Tasks

Train the model so that predicted values are as close as possible to actual values

- In regression problems, the model predicts continuous values (e.g., price, temperature, demand), not class probabilities.
- Maximum Likelihood Estimation (MLE) aims to:
  - Find the best model parameters (weights)
  - Such that the predicted values are as close as possible to the actual values

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MLE - Regression Tasks

How is this achieved?

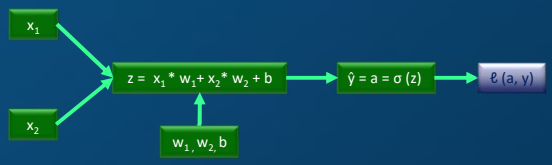
- MLE leads to minimizing the Mean Squared Error (MSE)
  - $MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$
  - y_i = actual value
  - $\hat{y}_i$ = predicted value
- Lower MSE = better predictions

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Understanding the Loss Function

- A loss function measures how well a model's predictions match the actual values.
- It compares:
  - Predicted output (ŷ)
  - True output (y)



- The difference between them is called the error
- The objective of training is to:
  - Minimize the loss (error)
  - Which means improving predictions step by step
- Lower loss = better model performance

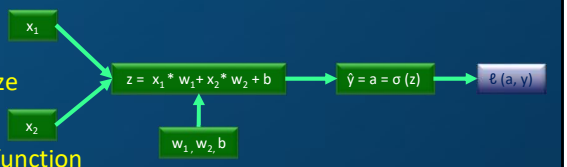
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Loss Function & Maximum Likelihood (MLE)

- Aim to maximize the probability of observed data
- This naturally leads to defining a loss function to minimize
- Maximizing likelihood = Minimizing an appropriate loss function



- Cross-Entropy Loss (Binary Classification)
  - Used when the model outputs probabilities (between 0 and 1)
  - $\ell(\hat{y}, y) = -[y * \log(\hat{y}) + (1 - y) * \log(1 - \hat{y})]$
- What Does It Do?
  - Cross-entropy measures how well:
  - The predicted probability (ŷ)
  - Matches the true label (y)

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Logistic Regression – Loss and Cost Function

Loss Function (for a single training example)

- For logistic regression (binary classification), the loss function is:
  - Loss($\hat{y}$, y) = − (y . log $\hat{y}$ + (1 − y) . log (1 − $\hat{y}$) for an instance or example
  - $y \in \{0, 1\}$→ true label
  - $\hat{y} \in (0, 1)$→ predicted probability
- If $y = 1$:
  - Loss reduces to: $-\log(\hat{y})$
- If $y = 0$:
  - Loss reduces to: $-\log(1 - \hat{y})$

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Logistic Regression – Loss and Cost Function

Cost Function (over the entire dataset)

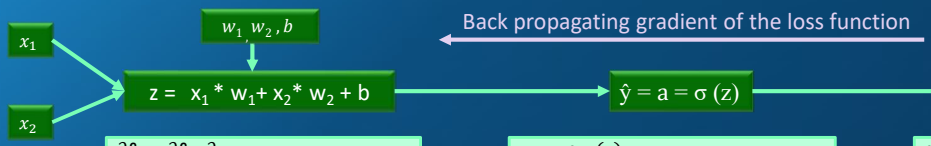
- The cost function is the average loss over all training examples:
  - $J(W, b) = \frac{1}{m}\sum_{i=1}^{m} \ell\left(\hat{y}^{(i)}, y^{(i)}\right)$
- Substituting the loss:
  - $J(W, b) = -\frac{1}{m}\sum_{i=1}^{m}\left[y^{(i)}\log\left(\hat{y}^{(i)}\right) + \left(1 - y^{(i)}\right)\log\left(1 - \hat{y}^{(i)}\right)\right]$
- The loss function measures error for one example
- The cost function measures average error over the entire dataset
- Training aims to:
  - Minimize $J(W, b)$ - Equivalent to Maximum Likelihood Estimation (MLE)

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Forward and Back Propagation



$$\frac{\partial \ell}{\partial z} = \frac{\partial \ell}{\partial a} \cdot \frac{\partial a}{\partial z}$$
$$= \{ -\frac{y}{a} + \frac{1-y}{1-a} \} * a * (1-a)$$
$$= a - y$$

$$\frac{\partial a}{\partial z} = \frac{\partial \sigma(z)}{\partial z} = \sigma(z) . (1 - \sigma(z))$$
$$\frac{\partial a}{\partial z} = a . (1 - a)$$

$$\frac{\partial \ell}{\partial a} = -\frac{y}{a} + \frac{1-y}{1-a}$$

z = X * W + b

ŷ = a = σ (z)

$\sigma(z) = \frac{1}{1+e^{-z}}$

ℓ(a, y) = - [y * log(a) + (1-y) * log(1-a)]

$\Rightarrow$

$$\frac{\partial \ell}{\partial w1} = x_1 . \frac{\partial \ell}{\partial z} = x_1 . (a-y)$$
$$\frac{\partial \ell}{\partial w2} = x_2 . \frac{\partial \ell}{\partial z} = x_2 . (a-y)$$
$$\frac{\partial \ell}{\partial b} = \frac{\partial \ell}{\partial z} = (a-y)$$

$\Rightarrow$

$$w_1 = w_1 - \alpha * \frac{\partial \ell}{\partial w_1} = w_1 - \alpha * x_1 * (a-y)$$
$$w_2 = w_2 - \alpha * \frac{\partial \ell}{\partial w_2} = w_2 - \alpha * x_2 * (a-y)$$
$$b = b - \alpha * \frac{\partial \ell}{\partial b} = b - \alpha * (a-y)$$

Where α is learning rate. The cost function is

$$J(W, b) = \frac{1}{m} * (\Sigma \ell (a, y))$$

Hence $\frac{\partial J}{\partial w_1} = \frac{1}{m} * (\Sigma \frac{\partial \ell (a, y)}{\partial w_1})$

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So where are the hidden layers!!!

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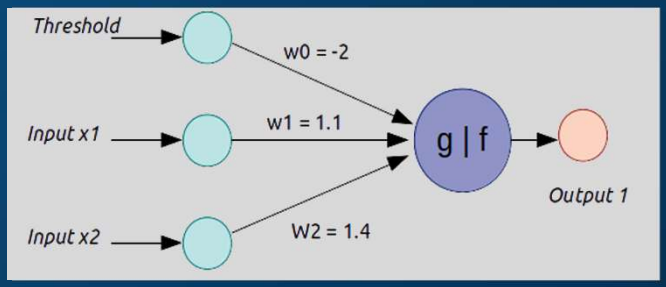
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Hidden Layers

id	Threshold	Team Members		Ground	
	x0	x1	w1	x2	w2
1	-2.00	1.00	1.10	1.00	1.40
2	-2.00	1.00	1.10	0.83	1.40
3	-2.00	1.00	1.10	0.67	1.40
4	-2.00	0.44	1.10	1.00	1.40
5	-2.00	0.22	1.10	0.00	1.40
6	-2.00	0.00	1.10	1.00	1.40

So far, we have considered only two features and one Threshold



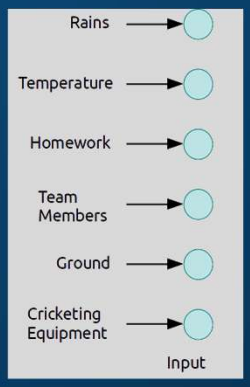
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Hidden Layers

id	Rains	Temp	Homework	Team Members	Equipment	Ground	Played
1	0.00	0.00	1.00	1.00	0.00	1.00	1
2	0.00	0.81	1.00	1.00	1.00	0.83	1
3	0.00	0.75	1.00	1.00	1.00	0.67	1
4	-0.17	0.69	1.00	0.44	1.00	1.00	0
5	-0.67	0.94	0.00	0.22	1.00	0.00	0
6	-1.00	1.00	0.00	0.00	0.00	1.00	0



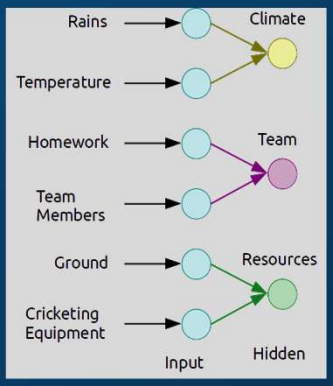
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Hidden Layers

id	Rains	Temp	Homework	Team Members	Equipment	Ground	Played
1	0.00	0.00	1.00	1.00	0.00	1.00	1
2	0.00	0.81	1.00	1.00	1.00	0.83	1
3	0.00	0.75	1.00	1.00	1.00	0.67	1
4	-0.17	0.69	1.00	0.44	1.00	1.00	0
5	-0.67	0.94	0.00	0.22	1.00	0.00	0
6	-1.00	1.00	0.00	0.00	0.00	1.00	0



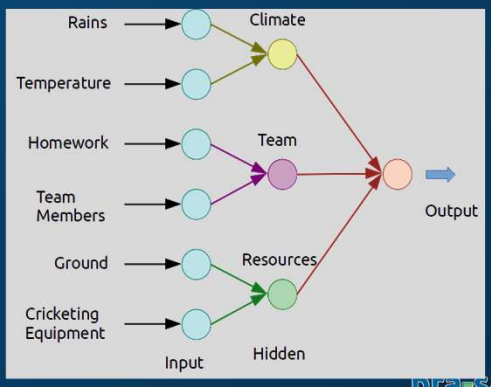
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Hidden Layers

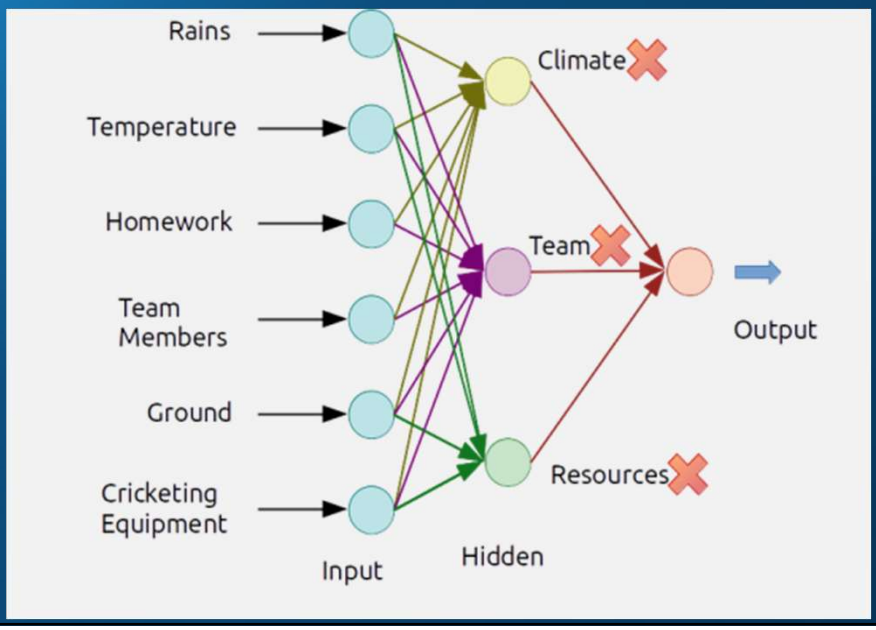
id	Rains	Temp	Homework	Team Members	Equipment	Ground	Played
1	0.00	0.00	1.00	1.00	0.00	1.00	1
2	0.00	0.81	1.00	1.00	1.00	0.83	1
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4	-0.17	0.69	1.00	0.44	1.00	1.00	0
5	-0.67	0.94	0.00	0.22	1.00	0.00	0
6	-1.00	1.00	0.00	0.00	0.00	1.00	0



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41 Hidden Layers



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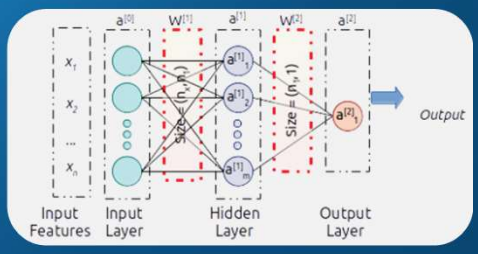
42 Hidden Layers



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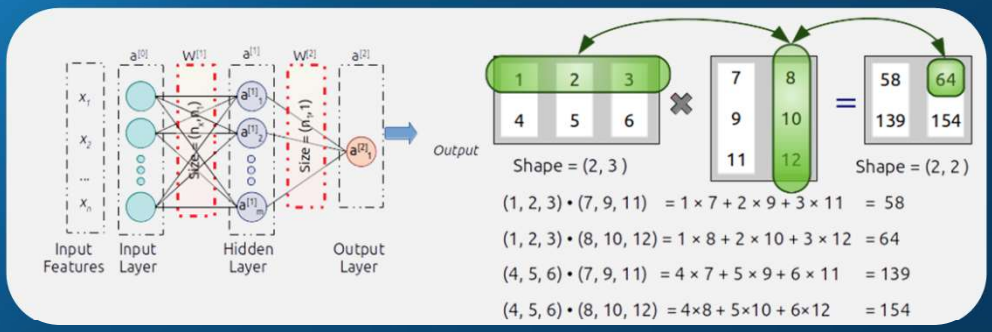
43 Two Major Conventions



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44 Two Major Conventions

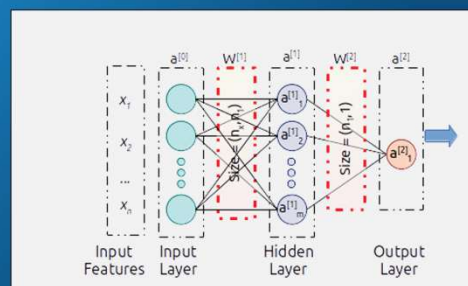


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Two Major Conventions



Output

1	2	3
4	5	6

Shape = (2, 3)

×

7	8
9	10
11	12

=

58	64
139	154

Shape = (2, 2)

(1, 2, 3) • (7, 9, 11) = 1 × 7 + 2 × 9 + 3 × 11 = 58

(1, 2, 3) • (8, 10, 12) = 1 × 8 + 2 × 10 + 3 × 12 = 64

(4, 5, 6) • (7, 9, 11) = 4 × 7 + 5 × 9 + 6 × 11 = 139

(4, 5, 6) • (8, 10, 12) = 4×8 + 5×10 + 6×12 = 154

X = | $X^{(1)}$ | $X^{(2)}$ | ... | $X^{(m)}$ | (n_x features)

m samples

X.shape = (n_x , m)

y = | $y^{(1)}$ | $y^{(2)}$ | ... | $y^{(m)}$ |

m samples

y.shape = (1 , m)

X = | -- $X^{(1)}$ -- | -- $X^{(2)}$ -- | -- ... -- | -- $X^{(m)}$ -- | (m samples)

n_x features

X.shape = (m, n_x)

y = | $y^{(1)}$ | $y^{(2)}$ | ... | $y^{(m)}$ | (m samples)

y.shape = (m, 1)

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Two M



Voilà !! Its time to update the resume!

X =

y =

64
154

(2, 2)

(m, 1)

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47

Reflect…

- Which statement correctly describes the relationship between MLE and cross-entropy in binary classification?
  - A. Cross-entropy loss is independent of MLE
  - B. Minimizing cross-entropy is equivalent to maximizing log-likelihood under a Bernoulli model
  - C. Cross-entropy maximizes prediction error
  - D. MLE minimizes squared error for classification
- Answer: B
- Why is the Bernoulli distribution used in logistic regression?
  - A. Because the model outputs continuous values
  - B. Because the target variable is binary and modeled via a probability parameter
  - C. Because it simplifies gradient computation
  - D. Because it prevents overfitting
- Answer: B

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Reflect…

- Why is the logarithm applied to the likelihood function in MLE?
  - A. To improve prediction accuracy
  - B. To convert products into sums and avoid numerical underflow during computation
  - C. To normalize output probabilities
  - D. To reduce model variance
- Answer: B
- What happens to cross-entropy loss when the predicted probability of the true class approaches 0?
  - A. It approaches 0
  - B. It remains constant
  - C. It diverges toward infinity
  - D. It becomes negative
- Answer: C

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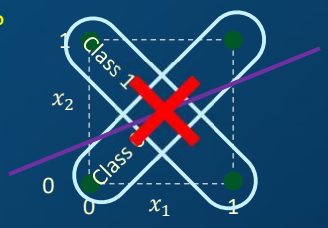
49 Reflect…

- What does the gradient of the loss function represent in gradient descent?
  - A. The optimal parameter values
  - B. The direction of steepest increase of the loss
  - C. The direction of steepest decrease of the loss
  - D. The second-order curvature of the loss
- Answer: B
- Why is cross-entropy preferred over MSE in logistic regression?
  - A. MSE cannot be applied to probabilistic outputs
  - B. MSE leads to non-convex loss in logistic models
  - C. MSE can produce smaller gradients when combined with sigmoid, slowing learning
  - D. MSE cannot handle binary targets
- Answer: C

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50 Reflect…

- Why can a single-layer neural network not represent the XOR function?
  - A. It lacks sufficient training data
  - B. It cannot apply activation functions
  - C. It can only model linearly separable decision boundaries
  - D. It cannot compute weighted sums



- Answer: C
- Which expression correctly represents forward propagation in layer l?
  - A. $a^{[l]} = W^{[l]} + b^{[l]}$
  - B. $a^{[l]} = \sigma\left(W^{[l]}a^{[l-1]} + b^{[l]}\right)$
  - C. $a^{[l]} = a^{[l-1]} - \nabla J$
  - D. $a^{[l]} = \log\left(W^{[l]}\right)$
- Answer: B

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Reflect…

- Why is the loss averaged over m examples in the cost function?
  - A. To reduce computational complexity
  - B. To normalize the loss and make it independent of dataset size
  - C. To prevent overfitting
  - D. To enforce convexity
- ✅ Answer: B
- In logistic regression, what does the output $\hat{y}$ represent?
  - A. A predicted class label
  - B. The expected loss value
  - C. The probability that y=1 given input x
  - D. The gradient of the loss function
- ✅ Answer: C

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Reflect…

- In forward propagation, what happens at each hidden layer?
  - A. The model checks for overfitting
  - B. The output is stored and not used further
  - C. A weighted sum is computed and passed through an activation function
  - D. The gradient is calculated for the next layer
- Answer: C. A weighted sum is computed and passed through an activation function
- In neural network notation, what $does\ \ a^{[l]}$ typically represent?
  - A. The weight matrix of layer l
  - B. The activation output of layer l
  - C. The cost function at layer l
  - D. The gradient with respect to weights of layer l
- Answer: B. The activation output of layer l

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53 Reflect…

- In forward propagation, what happens at each hidden layer?
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  - D. The gradient with respect to weights of layer l
- Answer: B. The activation output of layer l

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54 Reflect

- Why is optimization in logistic regression easier than in deep neural networks?
  - A. It uses fewer parameters
  - B. Its loss function is convex with respect to parameters
  - C. It does not require iterative optimization
  - D. It avoids local minima entirely

Note: D is tempting but incorrect (convex → no local minima issues, but wording is misleading)

- Answer: B
- Which statement is TRUE about minimizing cross-entropy loss?
  - A. It guarantees zero classification error
  - B. It encourages predicted probabilities to match the true data distribution
  - C. It maximizes Euclidean distance between predictions and labels
  - D. It removes the need for activation functions
- Answer: B

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Next Session - Coding Perceptron Model in Python

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ADDITIONAL MATERIAL

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Applications

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Bernoulli Likelihood → Logistic Regression Loss (MLE Derivation)

Modeling the Output (Bernoulli Distribution)

- In binary classification:
  - $y \in \{0, 1\}$
- The model predicts probability: $\hat{y} = P(y; =; 1 \mid xWb)$
- Assume $y \sim$ Bernoulli$(\hat{y})$
- The Bernoulli probability mass function:
  - $P(y; |; xWb) = \hat{y}^{y}(1-\hat{y})^{(1-y)}$
- This compact form handles both cases:
  - If $y = 1$: $P = \hat{y}$
  - If $y = 0$: $P = 1 - \hat{y}$

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59 Bernoulli Likelihood → Logistic Regression Loss (MLE Derivation)

Likelihood for Entire Dataset

- For m independent training examples:
  - $L(W, b) = \prod_{i=1}^{m} (\hat{y}^{(i)})^{y^{(i)}} (1 - \hat{y}^{(i)})^{(1-y^{(i)})}$
- This is the likelihood function (probability of observing the data)
- Log-Likelihood (to simplify computation)
  - Powers are scary, so take log
  - $\log L(W, b) = \sum_{i=1}^{m} [y^{(i)} \log(\hat{y}^{(i)}) + (1 - y^{(i)}) \log(1 - \hat{y}^{(i)})]$
- Convert to Minimization Problem
- In optimization, we usually minimize, so take negative:
  - $J(W, b) = -\frac{1}{m} \sum_{i=1}^{m} [y^{(i)} \log(\hat{y}^{(i)}) + (1 - y^{(i)}) \log(1 - \hat{y}^{(i)})]$

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60 Bernoulli Likelihood → Logistic Regression Loss (MLE Derivation)

Logistic regression loss = Negative log-likelihood of a Bernoulli model

- Start with **probability model (Bernoulli)**
- Apply **MLE (maximize likelihood)**
- Take **log for simplicity**
- Convert to **minimization**
  - Result = **Cross-Entropy Loss used in Logistic Regression**
- **Intuition**
  - If the model assigns **high probability to correct class** → likelihood increases
  - If it assigns **low probability** → likelihood drops sharply
  - Hence, loss penalizes **confident wrong predictions heavily**

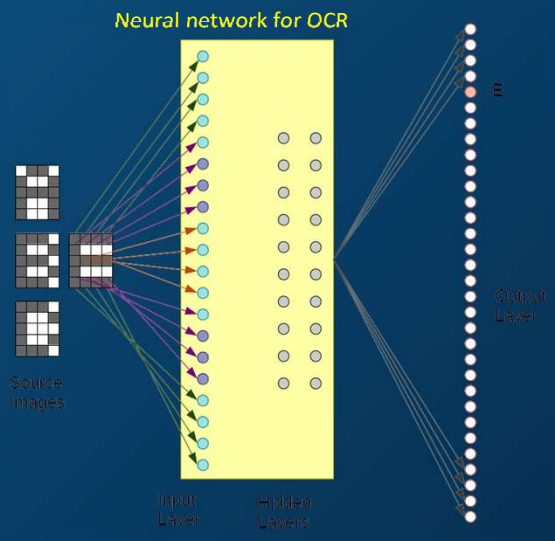
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Applications

- The properties of neural networks define where they are useful
- Typical Network
  - Can learn complex mappings from inputs to outputs, based solely on samples
  - Difficult to analyse
  - Firm predictions about neural network behaviour difficult;
    - Unsuitable for safety-critical applications.
  - Require limited understanding from trainer, who can be guided by heuristics



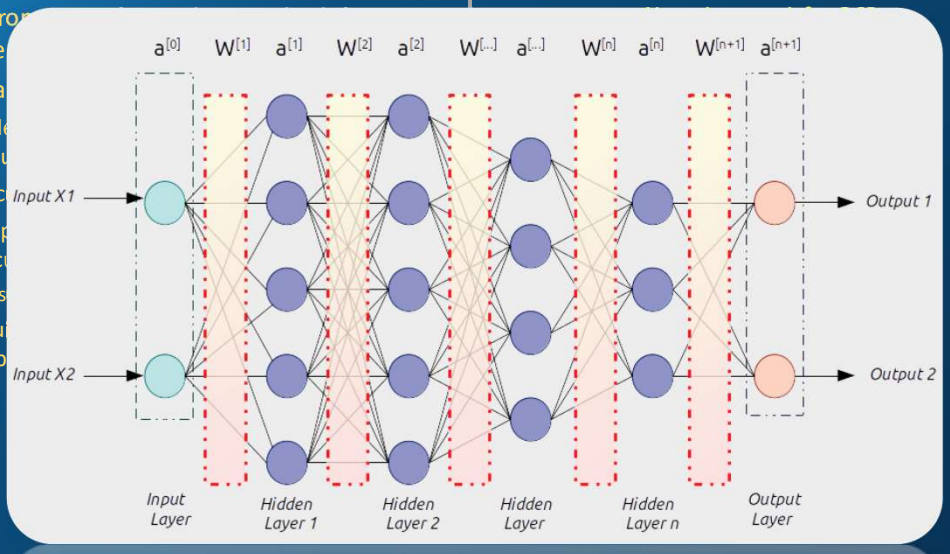
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Applications

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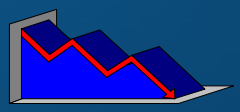
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Applications

- Stock market prediction
  - "Technical trading" refers to trading based solely on known statistical parameters; e.g. previous price
  - Neural networks have been used to attempt to predict changes in prices.
  - Difficult to assess success or otherwise
    - Since companies using these techniques are reluctant to disclose information.



- Mortgage assessment
  - Assess risk of lending to an individual
  - Difficult to decide on marginal cases
  - Neural networks have been trained to make decisions, based upon the opinions of expert underwriters
  - Neural network produced a 12% reduction in delinquencies compared with human experts



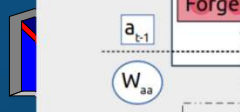
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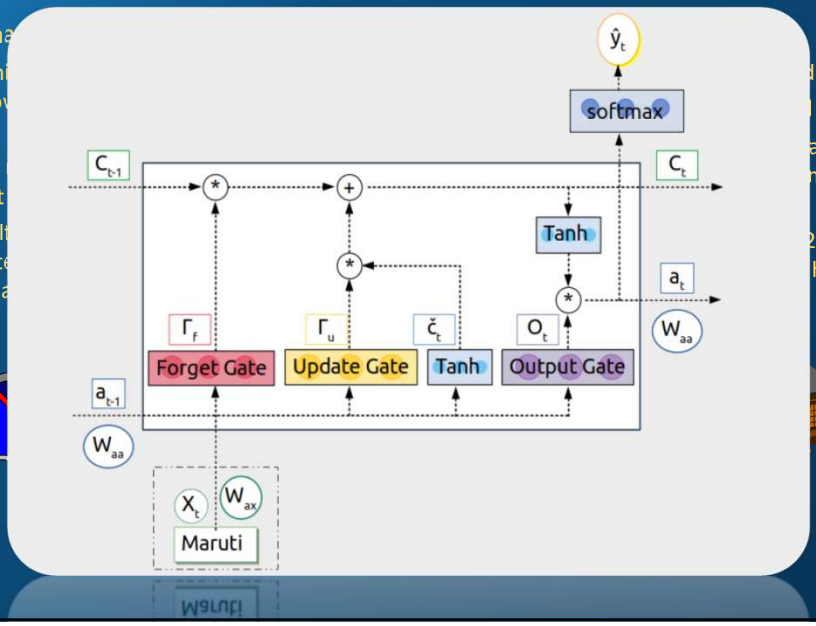
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Applications

- Stock ma
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Applications

- ALVINN: Autonomous Land Vehicle In a Neural Network

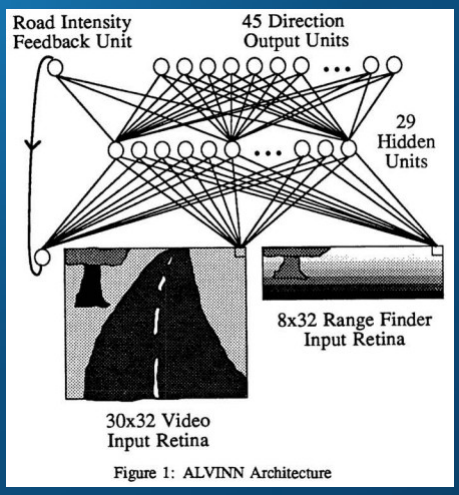
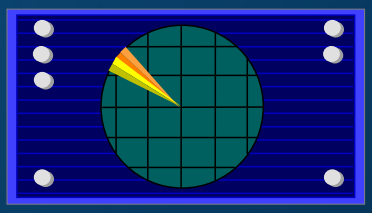


Figure 1: ALVINN Architecture

- Sonar target recognition
  - Distinguish mines from rocks on sea-bed
  - The neural network is provided with a large number of parameters which are extracted from the sonar signal.
  - The training set consists of sets of signals from rocks and mines.



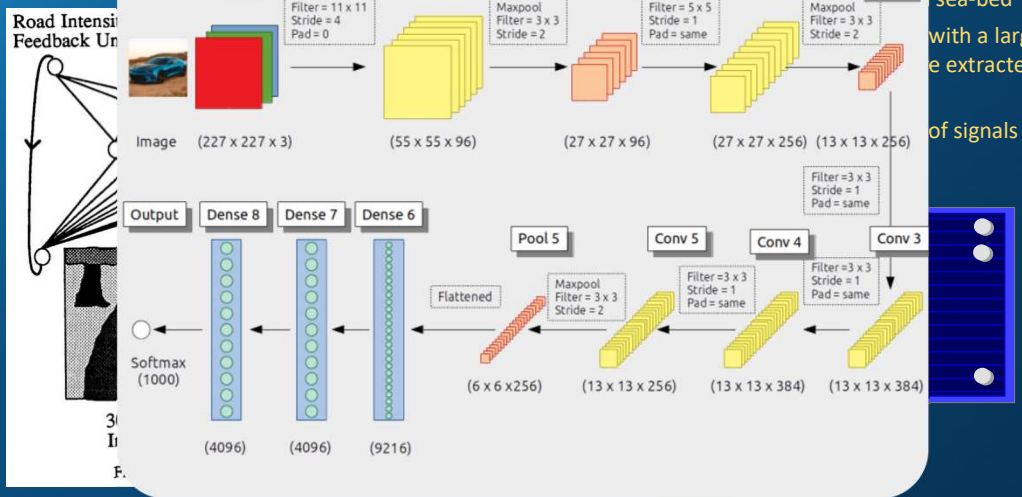
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Applications

- ALVINN: A Network



sea-bed
with a large
e extracted from

of signals from

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Applications

- Engine management
  - The behavior of a car engine is influenced by a large number of parameters
    - temperature at various points
    - fuel/air mixture
    - lubricant viscosity.
  - Major companies have used neural networks to dynamically tune an engine depending on current settings



- Signature recognition
  - Each person's signature is different.
  - There are structural similarities which are difficult to quantify.
  - Recognizes signatures to a high level of accuracy.
  - Considers speed in addition to gross shape
  - Makes forgery even more difficult.



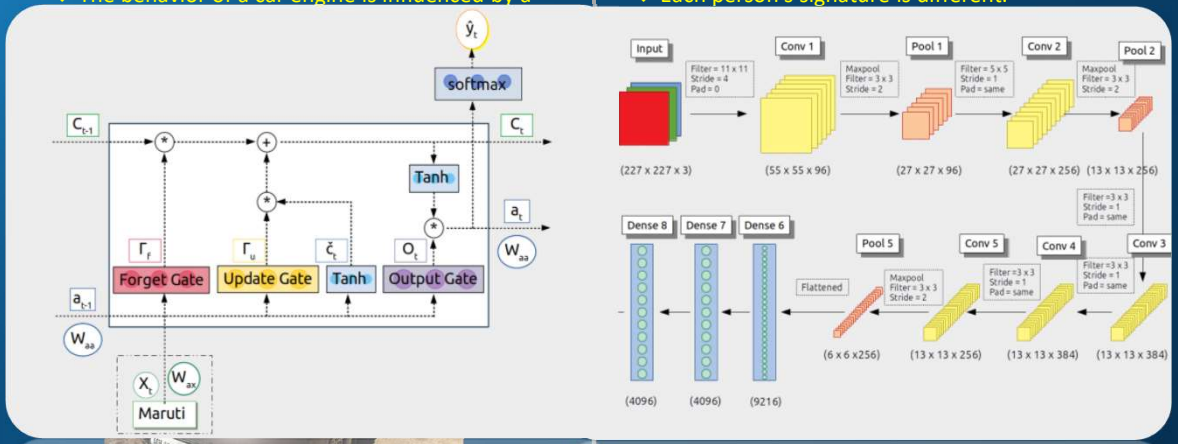
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Applications

- Engine management
  - The behavior of a car engine is influenced by a



- Signature recognition
  - Each person's signature is different.



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Derivation of Sigmoid

$$
\begin{aligned}
\partial a &= \partial\sigma(z) \\
&= \frac{\partial}{\partial z}\left[\frac{1}{1+e^{-z}}\right] \\
&= \frac{\partial}{\partial z}\left(1+e^{-z}\right)^{-1} \\
&= -(1+e^{-z})^{-2}(-e^{-z}) \\
&= \frac{e^{-z}}{(1+e^{-z})^2} \\
&= \frac{1}{1+e^{-z}} \circ \frac{e^{-z}}{1+e^{-z}} \\
&= \frac{1}{1+e^{-z}} \circ \frac{(1+e^{-z})-1}{1+e^{-z}} \\
&= \frac{1}{1+e^{-z}} \circ \left[\frac{1+e^{-z}}{1+e^{-z}} - \frac{1}{1+e^{-z}}\right] \\
&= \frac{1}{1+e^{-z}} \circ \left[1 - \frac{1}{1+e^{-z}}\right] \\
&= \sigma(z) \circ (1-\sigma(z)) \\
&= a \circ (1-a)
\end{aligned}
$$

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